1 साल का High-level Objective (North Star)

1. Technical Mastery: Automation, AI automation, Computer Vision, LLM engineering, Multi-Agent patterns, MLOps, production-grade deployment.

2. Tangible Deliverables in 1 year: 3 production-capable projects (deployed APIs), 1 mini-SaaS or paid tool, GitHub portfolio, résumé optimized for MAANG roles, interview-ready with system design + ML design, 3–5 client engagements or paying customers.

3. Outcome: Market-positioned as "senior AI/ML engineer — product + infra + LLMs + CV" and have a path to 50L/year via product + clients + scaled

services.

Tech Stack (focus)

- **Language: Python (main)**
- **Deep Learning: PyTorch (primary), TensorFlow (optional)**
- **LLMs & tools: Hugging Face transformers, embeddings, LangChain (agent patterns)**
- **CV: YOLO (Ultralytics), Detectron2 OR PyTorch custom, OpenCV for preprocessing**
- **MLOps / infra: Docker, Kubernetes (basic), CI/CD (GitHub Actions), MLflow**

/ Weights & Biases, Seldon/BentoML/FastAPI

- **Orchestration: Airflow (or Prefect) for pipelines**
- **Databases & infra: PostgreSQL/Redis, object storage S3 (or S3-compatible), cheap VPS / GCP/Azure credits for deployment**
- **Observability: logs + metrics (Prometheus/Grafana or W&B metrics + alerts)**
- **Basics: Linux, bash, REST, gRPC**

Monthly Plan — Detailed (Copy–paste actionable)

Month 0 — Prep (Week 0) — Set the foundation (1 week)

Deliverables:

- **Create GitHub account with professional README.**
- **Create a one-page portfolio (Notion/GitHub Pages).**
- **Set up Python dev env (venv/conda), Docker, Git, and VSCode.**
- **Time: 4–6 hours.**

Why: Smooth dev loop from day 1.

Month 1 — Fundamentals & Clean Coding (Weeks 1–4)

Goal: Solidify ML + DL foundations and reproducible code habits.

Weekly plan:

- **Weekdays: 1 hr theory (linear algebra, probability quick refresh), practical snippets.**
- **Weekend deep-dive: 6–8 hrs — implement from scratch a small neural net in PyTorch.**

Deliverables:

- **Short notebook: “My PyTorch starter” (train small CNN on CIFAR-10).**
- **Repo with reproducible README + training script + Dockerfile.**

Why: Employers/MAANG expect clean, reproducible experiments.

Month 2 — Computer Vision Core (Weeks 5–8)

Goal: Core CV skills — detection & segmentation.

Weekly plan:

- **Learn transfer learning, object detection basics (anchor boxes, NMS).**

- **Hands-on: Fine-tune a YOLOv8/Ultralytics model on a small custom dataset (10–20 classes or domain-specific).**

Deliverables:

- **Project: “Custom Object Detector” — training script, inference API (FastAPI),** Dockerized.

- Demo video (1–2 min) showing detection on sample images.

Metric: Get mAP > baseline on small dataset; API latency <500ms for 1 request.

Month 3 — Advanced CV & Robustness (Weeks 9–12)

Goal: Make CV models production-ready and robust.

Tasks:

- Data augmentation pipeline, handling occlusion/misalignment, QA tests.
- Add unit tests, model validation harness, and CI to training repo.

- **Add monitoring hooks (log validation metrics to W&B or logs).**

Deliverables:

- **Improved detector with augmentation & test suite; CI auto-run on push.**
- **Short doc: "How to reproduce my CV model".**

Why: Production-readiness separates junior from senior.

Month 4 — LLM Foundations & Embeddings (Weeks 13–16)

Goal: Understand LLM internals, embeddings, retrieval-augmented generation (RAG).

Tasks:

- **Learn tokenization, attention at high level, embeddings use-cases.**
- **Build an embeddings-based Q&A: use local open embeddings (or HF) + vector store (FAISS).**
- **Create a FastAPI endpoint that answers queries from a document set.**

Deliverables:

- **Project: “Docs Q&A” — upload doc -> create embeddings -> query API.**
- **Blog post or LinkedIn thread explaining RAG in simple terms.**

Metric: Query latency <1s (for small dataset), accuracy acceptable.

Month 5 — LLM Fine-tuning & Prompt Engineering (Weeks 17–20)

Goal: Fine-tune or instruction-tune small LLMs; master prompt engineering and evaluation.

Tasks:

- **Fine-tune a small open model on domain-specific data (or use parameter-efficient finetuning like LoRA).**

- **Build prompt templates + evaluation suite (automated metrics + human check).**

Deliverables:

- **Repo: “Customized LLM for X” with finetune scripts, LoRA config,**

evaluation notebooks.

- **Demo: Telegram/Slack bot or FastAPI endpoint that uses fine-tuned model.**

Why: Real-world LLM jobs value fine-tuning + evaluation.

Month 6 — Agents & Multi-Agent Patterns (Weeks 21–24)

Goal: Build and orchestrate simple agent workflows (LangChain/agent frameworks).

Tasks:

- **Implement a worker-manager pattern: manager issues subtasks, workers call LLMs + tools (search, code-execution, API call).**

- **Add guardrails: rate-limits, tool validation, human-in-loop thresholds.**

Deliverables:

- **Project: “Mini Agent Orchestrator” (manager UI optional) + docs.**
- **Stress test: 50 automated runs, log behavior, error recovery.**

Metric: System recovers gracefully from 90% of simulated failures.

Month 7 — MLOps Fundamentals (Weeks 25–28)

Goal: Learn CI/CD for ML, experiment tracking, reproducibility, model registry.

Tasks:

- Integrate MLflow/W&B for experiment tracking, version the model.

- Build a simple CI pipeline: tests, lint, train-if-needed, build Docker image, push.

- Put model into a registry and tag releases.

Deliverables:

- End-to-end CI pipeline in GitHub Actions that returns a deployable image.

- Short SOP doc: “Deploying and versioning a model”.

Why: MLOps is required for production roles.

Month 8 — Deployment & Scaling (Weeks 29–32)

Goal: Production deployment (Docker, Kubernetes basics, autoscaling) and observability.

Tasks:

- Deploy one project to a cloud VM or k8s cluster (minikube or cheap cloud).
- Add logging, metrics (latency, error rate), and A/B model rollout.
- Configure simple alerting.

Deliverables:

- Live deployed endpoint (testable) + monitoring dashboard screenshot.

- **Post: “How I deployed X with Docker + GitHub Actions”.**

Metric: Service handles 100 concurrent requests in simulation with <2s p95 latency.

Month 9 — Security, Privacy, Governance (Weeks 33–36)

Goal: Implement AI governance, safety, and compliance basics.

Tasks:

- **Add data privacy pipeline: data minimization, anonymization steps.**

- **Add input validation, content filtering, rate limiting.**

- **Put in human-in-loop thresholds and logs for auditing.**

Deliverables:

- **Governance checklist for each project (privacy, explainability, fallback).**

- **“Incident playbook” for model failure or hallucination.**

Why: Having governance knowledge is major differentiator.

Month 10 — Productize (Weeks 37–40)

Goal: Turn one project into a minimal SaaS MVP.

Tasks:

- Create pricing model, simple landing page, billing via Stripe (or simpler).
- Add onboarding flow, basic metrics (signup->trial->conversion).
- Set up simple analytics (who uses, frequency).

Deliverables:

- Live MVP with 1–2 paying users or commitment from potential clients.
- Marketing: producthunt launch plan or targeted outreach template.

Metric: 3–5 signups or 1 paying client (early traction).

Month 11 — Freelance & GTM (Weeks 41–44)

Goal: Start monetizing skills and products actively.

Tasks:

- **Apply to niche freelance gigs: pitch template, 1-pager technical spec, case studies.**
- **Reach out to target customers (cold email + LinkedIn), share demo.**
- **Convert first 1–3 paying clients for custom work or White-label SaaS.**

Deliverables:

- **3 client proposals sent, 1 signed contract.**

- **Revenue target month: ₹50k–₹2L.**

Why: Validates product-market fit and income.

Month 12 — Hire/Scale/MAANG Prep (Weeks 45–48)

Goal: Polish CV/LinkedIn, interview prep, and set path for scaling or MAANG apps.

Tasks:

- **System design for ML: study 5–7 real ML system design problems and write solutions.**
- **Mock interviews: data structures + ML systems + coding.**
- **Polish portfolio: case studies, architecture diagrams, costs and**

results.

Deliverables:

- **10 tailored job applications to target companies; 5 referrals via network.**
- **Final report: 1-year progress doc & 6-month scaling plan.**

Metric: 2–3 interviews at senior/lead AI roles or enough client pipeline to replace salary.

Weekly Routine (realistic when working full-time)

- **Mon–Fri: 1 hour learning (6–7 PM): 30m concept + 30m hands-on.**

- **Sat: 4–5 hours deep project work (training, debugging, writing docs).**
- **Sun: 6–8 hours — build features, deploy, write blog post / LinkedIn update, reach out to 5 people.**
- **15–30 minutes daily journal: what learned + blockers.**

Tip: Use time-blocking: “learning block” and “execution block” separate.

Portfolio & Hiring / Freelance Kit (what to have ready)

1. **GitHub: 3 main repos (CV detector, RAG Q&A, Agent orchestrator) with README, Dockerfile, tests.**

2. **Portfolio page: Project descriptions, architecture diagrams, cost/infra used, results (metrics).**

3. **Resume: Bullets quantifying impact ("reduced inference latency by 40%", "increased recall on OCR by X%").**

4. **LinkedIn: Weekly posts summarizing learnings & demos (video clips).**

5. **One-pager sales template for clients: problem → solution → deliverables → timeline → cost.**

6. **Pricing model: fixed-scope + maintenance retainer.**

Interview Prep (months 10–12)

- **Algorithms & DS: practice 3–4 weekly problems (LeetCode medium-level).**

- **ML system design: be able to present architecture for CV pipeline, RAG system, agent orchestration, with tradeoffs.**

- **Behavioral: STAR stories about projects (use metrics).**

- **Mock interviews: record and iterate.**

Freelancing → Scaling to Product (how to avoid trading hours for rupees)

1. **Early: take high-ticket freelance projects (₹50k–₹2L) that align with**

products.

2. Reuse: convert client custom work into features for the SaaS.

3. Subscription: move clients to monthly retainers for maintenance.

4. Automate onboarding & support with docs + chatbots.

Governance & Ethics Checklist (must-have for every project)

- Data: consent, minimal retention, anonymization.

- Bias: test model across demographics/datasets relevant to

problem.

- **Explainability: provide simple model behavior docs & confidence scores.**
- **Logging: store inputs, outputs, and decisions for audits (with retention policy).**
- **Human-in-loop: define thresholds where human review is mandatory.**
- **Safety tests: adversarial tests, prompt-injection tests for LLMs.**
- **Compliance: follow local laws (privacy, consumer protection) for customers.**

Include this checklist in repo README and product docs.

Measurable KPIs to track weekly/monthly

- **Weekly learning hours (target 12–18).**
- **Projects pushed to GitHub (target 1 per 1–2 months).**
- **Deployed endpoints (target 2 by month 6).**
- **Blog posts / LinkedIn posts (1 per 2 weeks).**
- **Number of outreach messages (20/week during GTM).**
- **Client revenue (target month 11–12 ₹50k–₹2L/month).**

- **Interview applications (10 targeted in months 10–12).**

Quick Starter Checklist — First 30 days (copy–paste)

1. **Setup dev env + Docker + GitHub repo.**
2. **Do Month1 deliverables (PyTorch starter).**
3. **Publish one LinkedIn post about what you built.**
4. **Schedule 2 deep-work weekend blocks.**

5. **Create one client outreach template.**

Common Pitfalls & How to Avoid

- **Trying to master everything at once → Focus on vertical: pick 1 CV project + 1 LLM project.**
- **No reproducibility → always dockerize + CI.**
- **Not networking → share progress weekly; reach out to people with specific asks.**
- **Skipping governance → later clients will reject for lack of auditability.**

Extra: 30-day Micro-challenges (for momentum)

- Day 1–7: Dockerize a simple model.
- Day 8–14: Deploy a FastAPI endpoint and hit it from Postman.
- Day 15–21: Add experiment tracking (W&B or MLflow).
- Day 22–30: Post a short demo video on LinkedIn + reach out to 10 people.

1-Year Roadmap to Become “Unbeatable AI Engineer + Solopreneur”

Phase 1 (0–3 Months): Foundation + Market Awareness

🎯 Goal → मजबूत नींव बनाना + समझना कि actual market में क्या demand है।

📚 Skills

- **Python Mastery** (DSA basics, OOP, async programming)
- **AI Core** →
    - Machine Learning (Scikit-learn, feature engineering, pipelines)
    - Deep Learning (PyTorch/TensorFlow basics, CNN, RNN, Transformers)
- **Computer Vision Starter** → OpenCV, YOLOv8 (custom dataset training), OCR.
- **LLM Starter** → HuggingFace basics, fine-tuning small models, LangChain.
- **MLOps Basics** → GitHub, Docker, FastAPI, simple CI/CD pipeline.

🛠 Execution

- Daily **2–3 hrs job learning + 2 hrs deep practice**.
- Build 2–3 mini projects:
    1. OCR pipeline (e.g., number plate reader).
    2. Simple chatbot using HuggingFace + LangChain.
    3. Custom YOLOv8 detector (own dataset).

💰 Freelance Prep

- Register on **Upwork, Toptal, Fiverr**.
- Create portfolio website (GitHub Pages/Notion → showcase projects).

Phase 2 (4–6 Months): Specialization + First Monetization

🎯 Goal → खुद को **AI + Automation specialist बनाना, पहली earning शुरू करना।**

📚 Skills

- **Advanced Computer Vision** → Object Tracking, Segmentation (Mask R-CNN, SAM).
- **LLM Engineering** → Prompt engineering, RAG pipelines (Vector DB: Pinecone/Weaviate), fine-tuning Llama-3.
- **AI Agents** → CrewAI, AutoGen, LangGraph.
- **MLOps Deep** → MLflow, DVC, monitoring.
- **Cloud Skills** → Azure ML, AWS Sagemaker, GCP VertexAI (basic deployment).

🛠 Execution

- Build **3 strong projects**:
    1. Parking system (YOLO + OCR + rule engine).
    2. RAG-powered chatbot with multiple agents.
    3. AI-powered automation script (email responder / report generator).

💰 Monetization

- Start bidding on niche projects → "OCR automation", "LLM chatbot", "Computer Vision system".
- First client target: $500–$1k project.

Phase 3 (7–9 Months): Scaling + Authority Building

🎯 Goal → Freelance reputation बनाना + खुद का system ready करना।

📚 Skills

- **AGI & Future Track** → Research papers (RLHF, Multi-Agent Systems, OpenAI papers).
- **Governance & Ethics** → Responsible AI, bias detection, GDPR/Indian IT laws.
- **Full-stack AI Product** → React + FastAPI + MongoDB + PyTorch/LLM.

🛠 Execution

- Build **flagship project** (end-to-end SaaS MVP):
  Example → “AI Resume Analyzer”, “AI Compliance Checker”, “AI-driven Traffic Analytics”.
- Start **LinkedIn + Twitter posting**: weekly technical threads + project showcase.

💰 Monetization

- Charge higher ($1.5k–$3k/project).
- Offer **subscription-based automation tools** (side SaaS).

Phase 4 (10–12 Months): Solopreneur Mode ON

🎯 Goal → Secure job + parallel income system.

📚 Skills

- Advance **MLOps** (Kubernetes, Kubeflow).
- Learn **business basics** (pricing, pitching, sales funnel).
- Create **personal AI agent system** (agents that code, deploy, test).

🛠 Execution

- Launch **paid tool/product** on ProductHunt, Gumroad, or IndieHackers.
- Setup **freelance agency profile** (not just individual).
- Contribute to open-source AI repo → builds credibility.

💰 Income Balance

- Job: ₹5–8LPA (or upgrade to ₹12–20LPA MAANG-track role).
- Freelance/SaaS: $2k–$5k/month.
- Start saving/investing 30% in mutual funds/crypto (future buffer).

🛡 Final Outcome after 12 Months

✔ Strong portfolio (CV + LLM + Agents + MLOps).
✔ Secure job + freelance clients + 1–2 SaaS tools live.
✔ Positioning: “AI Automation & Agent Specialist” (rare + high demand).
✔ Demand in MAANG-level interviews (AI infra/LLM teams).
✔ Financial runway + spiritual time secured.